

YEAR 10 SCIENCE (GENERAL)
CHEMICAL SCIENCE
TEST 2

Name: SOLUTIONS

Form: _____

Mark: 56

The atomic structure and properties of elements are used to organise them in the Periodic Table.

21

(Mult. choice Q 1 - 8, Part 2 Q 1 - 3)

Different types of chemical reactions are used to produce a range of products and can occur at different rates.

35

(Mult. choice Q 9 - 15, Part 2 Q 4 - 8)

Part 1: Multiple Choice

Write your choice for each question in the table below.

1. Which of the following all have **2 valence electrons**?

- (a) F, Cl, Br.
- (b) Li, Na, Al.
- (c) Cl, Al, C.
- (d) Be, Ca, Mg.

2. Which of the following statements concerning groups in the Periodic Table is **not correct**?

- (a) Groups contain either all metals or all non-metals.
- (b) Groups contain elements with similar chemical properties.
- (c) Groups are the vertical columns in the table.
- (d) Groups are numbered using the Roman numerals I to VIII.

3. The mass number of an atom is the total:

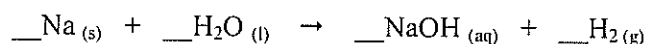
- (a) charge on the nucleus of the atom.
- (b) number of protons in the atom.
- (c) mass of the atom expressed in grams.
- (d) number of protons and neutrons in the atom.

4. The **valence electrons** are:

- (a) the number of total electrons an atom has.
- (b) the number of electrons in the atom's outer shell.
- (c) the number of electrons an atom has after it reacts.
- (d) the number of electrons an atom has before it reacts.

MULTIPLE CHOICE ANSWERS	
1	D
2	A
3	D
4	B
5	B
6	D
7	D
8	A
9	C
10	A
11	B
12	C
13	B
14	A or C
15	C

5. If the electron configuration of an atom is 2, 8, 2, to which group of the periodic table does it belong?
- Group I—alkali metals.
 - Group II—alkaline earth metals.
 - Group III.
 - Group VII—halogens.
6. Which of the following is the most important factor in determining the chemical properties of an element?
- The number of neutrons in the nucleus of its atoms.
 - The number of protons in the nucleus of its atoms.
 - The total number of electrons in its atoms.
 - The number of valence electrons in its atoms.
7. The electron configuration 2 8 4 is that of:
- neon.
 - nitrogen.
 - carbon.
 - silicon.
8. The atom represented by the symbol $^{52}_{24}\text{Cr}$ has:
- 24 protons, 28 neutrons and 24 electrons.
 - 28 protons, 24 neutrons and 28 electrons.
 - 52 protons, 52 neutrons and 52 electrons.
 - 24 protons, 52 neutrons and 24 electrons.
9. The formula for dinitrogen tetrahydride is:
- N_5H_2 .
 - N_2H_5 .
 - N_2H_4 .
 - N_4H_2 .
10. Which of the following word equations correctly explains an acid reacting with a metal?
- zinc + hydrochloric acid $\rightarrow$ zinc chloride + hydrogen
 - magnesium + sulphuric acid $\rightarrow$ magnesium chloride + hydrogen
 - magnesium + hydrochloric acid $\rightarrow$ carbon dioxide + hydrogen
 - zinc + hydrochloric acid $\rightarrow$ zinc chloride + carbon dioxide
11. The equation shown below is not balanced.



The coefficients required to balance this reaction, in order, are:

- 1, 2, 1, 1.
- 2, 2, 2, 1.
- 1, 1, 1, 2.
- 2, 2, 1, 1.

12. The test for determining whether a colourless, odourless gas is hydrogen is called the:
- glowing splint test.
 - pH test.
 - pop test.
 - limewater test.
13. Which of the following ionic formulae are **ALL** correct?
- NaI, NaO, Na₂O₄
 - MgO, MgCl₂, Mg₃(PO₄)₂
 - CaO₂, CaF, CaCl₂
 - Al₂O₃, Al₂Cl₃, AlF₃
14. H₂ is the formula for:
- a molecule.
 - a compound.
 - an element.
 - an atom.
15. The calcium ion carries a charge of 2+, the nitrate ion a charge of 1-. The correct formula for calcium nitrate is:
- CaNO₃.
 - Ca₂(NO₃).
 - Ca(NO₃)₂.
 - CaNO₃₂.

Part 2: Short Answer

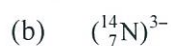
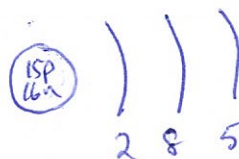
1. Complete the following table.

ELEMENT / ION	PROTON NUMBER	NEUTRON NUMBER	ELECTRON NUMBER
²⁴ ₁₂ Mg	12	12	12
(³⁵ ₁₇ Cl) ¹⁻	17	18	18
⁵⁶ ₂₆ Fe	26	30	26
(²⁷ ₁₃ Al) ³⁺	13	14	10
¹⁶ ₈ O	8	8	8

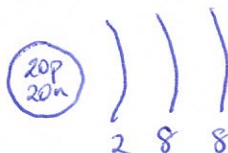
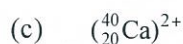
(1 mark each row)

(5)

2. Draw the **electron configuration** diagrams for the following species. Remember to show the constituent particles of the nucleus.



(Nucleus - 1 mark Shells - 1 mark)



(6)

3. Use the periodic table to predict the likely charges of the ions formed by the atoms of the following elements.

ELEMENT	CHARGE ON ION FORMED
Br	1-
Sr	2+
Se	2-
Fr	1+

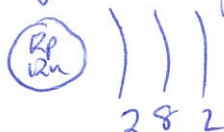
($\frac{1}{2}$ mark each)

(2)

4. The formula for magnesium chloride is MgCl_2 . This arrangement of ions is very stable and follows the Octet Rule.

- (a) Using Mg as an example, explain how it forms the Mg^{2+} ion and why it is so stable.

- Mg has the following electron configuration.



($\frac{1}{2}$)

- It has 2 valence electrons in the outer shell.

($\frac{1}{2}$)

- These electrons are donated.

($\frac{1}{2}$)

- Mg^{2+} ion has the same electron configuration as Ne, the nearest inert gas (∴ it is stable).

($\frac{1}{2}$)⁽²⁾

(b) Explain why the formula for magnesium chloride is MgCl_2 and not MgCl .

- Mg^{2+} has donated 2 electrons. ($\frac{1}{2}$)
- Cl^- has gained 1 electron. ($\frac{1}{2}$)
- There must be 2 Cl^- ions to use both donated electrons. (1).

(2)

5. Write the correct formulae for the following.

NAME	FORMULA
barium chloride	BaCl_2
dinitrogen tetroxide	N_2O_4
aluminium carbonate	$\text{Al}_2(\text{CO}_3)_3$
copper (II) hydroxide	$\text{Cu}(\text{OH})_2$
diphosphorus pentoxide	P_2O_5

NAME	FORMULA
sodium phosphate	Na_3PO_4
magnesium sulphide	MgS
carbon monoxide	CO
calcium fluoride	CaF_2
zinc sulphate	ZnSO_4

(1 mark each).

(10)

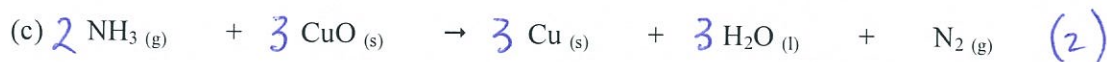
6. Ionic substances such as table salt (NaCl) won't conduct electricity when it is a solid in a pile, but it will conduct when it is dissolved into water. Explain why this is the case.

Solid - no free electrons moving or ions moving. (1)

Solution - ions are free to move and act as charge carriers. (1)

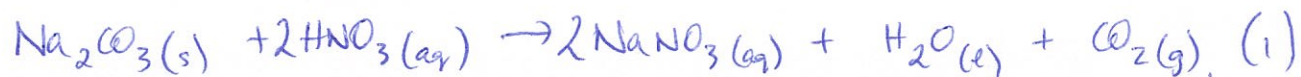
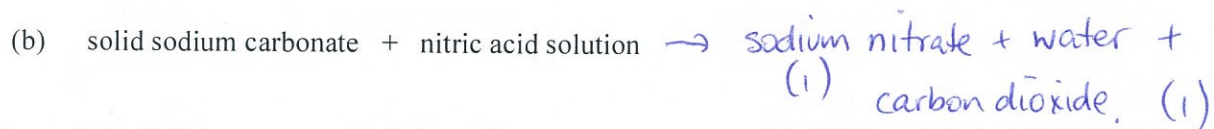
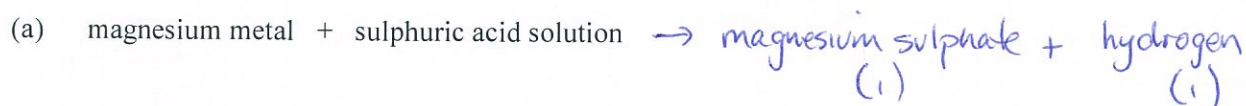
(2)

7. Balance the following equations.



(6)

8. Write **word equations** for each of the following, and then a **balanced equation** underneath each one.



(6)